FEI HAN

Robot Learning · Dynamics Modeling · Bioinspired Robotics

Email: hanfei@westlake.edu.cn | Website: hanfei-hz.github.io

Joint PhD Program · Expected completion June 2027

Website QR

Education

Westlake University and Zhejiang University
Joint PhD Program · Computer Science and Technology
Aug 2023 - Jun 2027 (expected) · Advisor: Prof. Dixia Fan · i4FSI Lab

University of Chinese Academy of Sciences · MEng · Sep 2019 - Jun 2022
Pattern Recognition and Intelligent Systems · Advisor: Xingang Zhao
Research at the State Key Laboratory of Robotics, Shenyang Institute of Automation, CAS.

Zhengzhou University · BEng
Measurement and Control Technology and Instruments · Sep 2015 - Jul 2019

Research Appointments

Pennsylvania State University · Visiting Scholar · Apr - Oct 2026

Westlake University · Research Assistant · Jul 2022 - Aug 2023

Research Contributions

Bioinspired aquatic actuator modeling and control[2, 9, 11] **(Dec 2023 - Present):** Measure forces and model hydrodynamics for learning-based control of webbed feet, fins and tails; contribute to REEF's real-fluid learning framework and platform, and HydroWrench's offline load reconstruction benchmark.

Quadruped swimming control and safe learning[1, 2, 10]**:** Use component hydrodynamic models for whole-robot gait optimization and hardware validation; transfer periodic gaits learned safely on a single leg to the robot. Study behavior planning and route tracking in simulation.

Early storm-petrel simulation research[3]**:** Contributed to air-water modeling, deep RL tuning and simulation analysis of pattering and sea-anchoring.

Deep-sea soft manipulation[6-8]**:** Contributed to NSFC Major Research Plan Integrated Project No. 92048302, studying robots that combine rigid, flexible and soft structures for operations in extreme environments. As a core student contributor to the soft-gripper work, led design, fabrication, sensing and control development, and guided junior students under the advisor's supervision.

Multilegged swimming (Apr 2026 - Present): Lead research on multilegged swimming, including reduced-order dynamics modeling, numerical simulation and experimental design. Collaborate with researchers from Harvard University, Carnegie Mellon University and other institutions; manuscript in preparation. | Baxi Chong group

Publications and Manuscripts

* Equal contribution. † Co-corresponding author. Applicant name in bold.

Published and Accepted

1. Xinyu Cui*, **Fei Han*†**, et al. Swimming Under Constraints: A Safe Reinforcement Learning Framework for Quadrupedal Bio-Inspired Propulsion.
ICRA · 2026 · Accepted

2. **Fei Han**, Pengming Guo, Hao Chen, et al. Learn to Swim: Data-Driven LSTM Hydrodynamic Model for Quadruped Robot Gait Optimization.
ICRA · 2025 · pp. 10261-10267 · Published

Publications and Manuscripts Continued

Published and Accepted Continued

3. Jiaqi Xue*, **Fei Han***, et al. Exploring Storm Petrel Pattering and Sea-Anchoring Using Deep Reinforcement Learning.
Bioinspiration & Biomimetics · 2023 · 18(6): 066016 · Published

4. Linlin Kang, Shiying Xiong, Buchen Wu, Chao Li, Mingming Ge, Dehan Yuan, **Fei Han**, Weicheng Cui. Force Decomposition on Flapping Flexible Plate via Impulse Theory and Dynamic Mode Decomposition.
Physics of Fluids · 2023 · 35(10): 101914 · Published

5. Linlin Kang, An-Kang Gao, **Fei Han**, et al. Propulsive Performance and Vortex Dynamics of Jellyfish-like Propulsion with Burst-and-Coast Strategy.
Physics of Fluids · 2023 · 35(9): 091904 · Published

6. **Fei Han**, Daohui Zhang, Xingang Zhao, et al. Design of a Composite-Cavity Bioinspired Soft Hand for Underwater Grasping.
Robot · 2023 · 45(2): 207-217 · Published · In Chinese; title translated

7. **Fei Han**, Daohui Zhang, Xingang Zhao. Human-Robot Interaction Control of a Soft Hand Based on a Curvature Prediction Model.
Automation & Instrumentation · 2022 · 37(7): 43-47 · Published · In Chinese; title translated

8. Yanxu Sun, **Fei Han**, Daohui Zhang, et al. An Underwater Soft Claw Based on Bionic Principle.
IEEE ROBIO · 2021 · Published

Under Review

9. **Fei Han***, Xinyu Cui*, Zhipeng Wang*, et al. Embracing Flow Unsteadiness: A High-Throughput Learning Platform Enables Vortex-Exploiting Bioinspired Propulsion.
Nature Mechanical Engineering · Under review · arXiv:2609.14254 (2026)

10. From Gaits to Routes: Execution-Aware Hybrid Behavior Planning for Quadrupedal Swimming with Experiment-Calibrated Hydrodynamics.
ICRA · 2027 · Under review · **First author**

11. HydroWrench: A Dataset and Benchmark for Dynamic Load Reconstruction in Bioinspired Aquatic Mechanisms.
ICLR · 2027 · Under review · **First author**

Academic Service and Campus Engagement

Westlake visits during IROS 2025: Core organizer and presenter for laboratory-hosted visits; welcomed academic and industry guests and introduced Westlake, the laboratory and its research.

Laboratory open days and science outreach: Hosted visits by primary and secondary school students and explained research platforms, bioinspired robotics and fluid mechanics.

Intelligent fluids conference: Volunteered in conference logistics and on-site service for an event hosted by the School of Engineering.

Undergraduate residential-college exchanges: Shared research experiences and reflections on postgraduate study as a graduate student.

Westlake University choir: Participated in rehearsals and university performances.

Research Groups



i4FSI Lab · i4fsi.lab.westlake.edu.cn



Baxi Chong group · sites.psu.edu/libr/